

ABDULLAH KHAWAJA

London, United Kingdom | +44 7501 754015 | abdullah_mujeeb97@hotmail.com
[LinkedIn](#) | [GitHub](#) | [Hugging Face](#) | [Substack](#) | [HackerNoon](#)

SKILLS

GenAI & LLMs:	Unsloth, LangChain, LangGraph, LangSmith, Ollama, OpenAI, Claude Code, Cursor, Gemini CLI, vLLM, RAG, embeddings, LLM post-training (SFT, DPO), fine-tuning (Gemma / Llama).
Machine Learning:	PyTorch, TensorFlow, Keras, Scikit-Learn, XGBoost, Vertex AI, BigQuery, A/B experimentation, model monitoring & drift detection, recommender systems (DeepFM, two-tower retrieval, neural rankers, LTR).
Development:	Python, SQL, Java, C++, React, TypeScript, Electron, FastAPI, Docker, Git, Redis / BullMQ.
Cloud & Tools:	Alibaba Cloud, AWS, GCP, Vercel, Hetzner, Postman, Tableau, Hugging Face, Sentence-Transformers.

WORK EXPERIENCE

Machine Learning Engineer | DPD Group

Nov 2023 - Present

- **Foundation models** - End-to-end design, training, and evaluation of the consumer-chatbot replacement model: ran a base-model bake-off across **Qwen 0.8B**, **Gemma 2B-IT**, and **Gemma 4B-IT**; fine-tuned the chosen Gemma 4B-IT model on **20K+** internal chatbot conversations with tool calls and response-time databook recall, then applied **DPO** from recorded internal feedback.
- **LLM serving** - Built the runtime router that chooses between the local model and **Gemini 3.5 Flash** using an Opus 4.8 LLM-as-Judge framework and custom benchmarks for emotional intelligence, helpfulness, tone, instruction-following, factuality, and reply quality; reduced projected serving cost from **~20K GBP** to **~4K GBP** per month (**~80%**).
- **Churn Prediction** - Architected a 6-model meta-ensemble that produced the strongest predictive performance across DPD Group, identifying **64%** of flagged customers as true churners and capturing **67%** of total churn cases; **\$650K** savings recorded so far.
- **Volume Forecasting** - Built a deep learning forecasting model (Mixture of Experts, RNN, WaveNet) beating a 12-year human benchmark by **+8%**; event-based lags improved seasonality across **80+** hubs.

Data Scientist (Contract) | Dastgyr (Remote)

Jun 2023 - Nov 2023

- **Recommendation** - Implemented SWING collaborative filtering from scratch for the Top For You module, driving **+1.44%** basket-size uplift and **+22%** average-order-value growth.
- **Strategy** - Defined the AI roadmap across sales forecasting and out-of-stock reduction; wrote the docs, experiment plans, and follow-ups needed to get the work shipped.

Senior Data Scientist - Recommendations | Daraz (Alibaba Group)

Oct 2020 - Feb 2023

- **Ranking models** - Implemented Daraz's first **DeepFM** product-ranking model, moving the rank stage beyond logistic regression with richer feature interactions and lifting click-through rate by **+5%**.
- **Campaign ranking** - Introduced an LTR re-rank stage for app campaign traffic, covering the banner-led entry points that made up roughly **40%** of app surface coverage; gave operators controlled trade-offs across click, add-to-cart, and order objectives, lifting conversion rate by **~34%** over baseline.
- **Product recall** - Built the Trending Now backend and Search Tips recall strategy; best Trending Now variant hit **~20% CTR** in multiple categories (**+46% CTR**, **+26% CR** vs single-category baseline) while Search Tips lifted Search CR by **~5%**.
- **Team leadership** - Led **5** data scientists across personalization, ranking, and recommendation for Daraz's **5** country ventures; owned A/B testing across **50+** experiments, interviewed candidates, and mentored new Recommendation team hires.

PROJECTS

[Sort Moments | GitHub](#)

- **Tech:** Python, PyQt6, RetinaFace, Buffalo Large (Embeddings).
- **Description:** A privacy-first desktop application for large-scale event photography (4,000+ images). Public site shows **407 downloads** and **2,501 visitors**. Runs a fully local RetinaFace + InsightFace pipeline using DirectML to detect and cluster faces, then organize solo and group shots by person without data leaving the device.
- **Features:** Offline processing, rapid image-size reduction, high-precision face clustering, manual refinement, and deduplication.

[WeaveSkip](#)

- **Tech:** Next.js, TypeScript, Prisma, PostgreSQL, Redis / BullMQ, NextAuth, OAuth, S3, Stripe, Zod.
- **Description:** A creator distribution OS that turns one article, RSS feed, or manual source into platform-native drafts for LinkedIn, X, Reddit, Facebook, Threads, and Substack Notes. Public site shows **5,344 visitors**, **3 signups**, **8 post batches**, **100% kept without edits**, and **2 finished posts**.
- **Features:** Source ingestion, intent / keyword / subreddit intelligence, BYOK LLM config with encrypted keys, human review, guardrails, queue-based scheduling / publishing, and Medium crosspost support.

[NeighbourFit](#)

- **Tech:** React, TypeScript, FastAPI, PostgreSQL / PostGIS, Mapbox, OpenAI, scikit-learn.
- **Description:** A UK rental postcode checker and area-fit tool. Public counter shows **136 postcode lookups**; Map Intelligence and UK postcode checks are open while profile, saved searches, and agentic rental discovery stay in private beta.
- **Features:** Commute and amenity context, geospatial scoring, maps, source-aware caveats, saved-list flow, and fit breakdowns that show context rather than pretending to give a verdict.

[Hugging Face Model Releases](#)

- **Tech:** Gemma, Qwen, LoRA / fine-tuning, ablation, distillation, GGUF quantization.
- **Description:** Public model releases exploring small-model adaptation, ablation experiments, Opus-distilled outputs, and deployment-ready GGUF packaging.
- **Focus:** Hands-on LLM training, model surgery, quantization, and open-source distribution across multiple experimental releases.

[Navigating Noise | HackerNoon](#)

- **Focus:** Technical writing on ML systems, AI tools, benchmarks, interviews, and product lessons for engineers and builders.
- **Description:** Converts noisy AI developments into practical mental models: what to follow, how to prepare, and how to judge tools that claim to change workflows.

EDUCATION

Master of Science in Data Science (Distinction) | University of Essex, UK

Oct 2022 - Oct 2023

Bachelor of Science in Computer Science | FAST-NU, Pakistan

Aug 2016 - Sep 2020